

Kugel: $V = \frac{4}{3} \cdot \pi \cdot R^3$ $A = 4 \cdot \pi \cdot R^2$
 Zylinder: $V = \pi \cdot R^2 \cdot h$ $A_m = 2 \cdot \pi \cdot R \cdot h$
 Kreis: $A = \pi \cdot R^2$ $U = 2 \cdot \pi \cdot R$

Kugelladung (Gauß) (elektrisches Feld in Abhängigkeit von r)

... Ladungsdichte

R... Radius d. Kugel (konstant)

F... Radius zum einsetzen

$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{eing}}}{\epsilon_0} \rightarrow E(r) \cdot A = \frac{Q_{\text{eing}}}{\epsilon_0}$

$Q_{\text{eing}}(r) = p \cdot V(r)$ $p = \frac{Q_{\text{Ges}}}{V_{\text{Ges}}}$ $\text{Ges. Ladung} = \frac{Q}{V(R)}$

Linienladung: (p) von unendl. Linienladung auf z-Achse

p... Linienladungsdichte

$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{eing}}}{\epsilon_0} \rightarrow E(r) \cdot A_m = \frac{Q_{\text{eing}}}{\epsilon_0}$ $Q_{\text{eing}} = pL \cdot l$

$E = -\nabla\phi$ $\phi(r) = -\frac{pL}{2\pi\epsilon_0} \cdot \ln\left(\frac{r}{r_0}\right)$

Geladene Kreisscheibe (p u. E entlang z):

p... Linienladungsdichte

p... oberflächendichte

p... oberflächendichte

$\phi = \frac{1}{4\pi\epsilon_0} \cdot \frac{q}{r} \rightarrow \phi = \frac{1}{4\pi\epsilon_0} \cdot \frac{dq}{\text{Abstand}} \rightarrow \phi = \frac{1}{4\pi\epsilon_0} \cdot \frac{1}{\sqrt{r^2 + z^2}}$

$dq = p \cdot dA$ $dA = 2\pi r \cdot dr$ $\vec{E} = \frac{d\phi}{dr}$

$\phi = \frac{1}{2\epsilon_0} \int_0^z \frac{1}{\sqrt{r^2 + z^2}} dr$ $\vec{E} = \frac{d\phi}{dz}$

Laplace Zylinderkoordinaten:

$\nabla^2\phi = \frac{1}{\rho} \frac{\partial}{\partial\rho} \left(\rho \frac{\partial\phi}{\partial\rho} \right) + \frac{1}{\rho^2} \frac{\partial}{\partial\varphi} \left(\rho \frac{\partial\phi}{\partial\varphi} \right) + \frac{\partial^2\phi}{\partial z^2}$

Wenn: $\phi = \phi(\rho) \rightarrow \frac{1}{\rho} \frac{\partial}{\partial\rho} \left(\rho \frac{\partial\phi}{\partial\rho} \right) = 0$

$\phi(\rho) = C_1 \cdot \ln(\rho) + C_2$

Laplace Kugelkoordinaten:

$\nabla^2\phi = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial\phi}{\partial r} \right) + \frac{1}{r^2 \sin(\theta)} \frac{\partial}{\partial\theta} \left(\sin(\theta) \frac{\partial\phi}{\partial\theta} \right) + \frac{1}{r^2 \sin^2(\theta)} \frac{\partial^2\phi}{\partial\varphi^2}$

Wenn: $\phi = \phi(r) \rightarrow \nabla^2\phi = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial\phi}{\partial r} \right) = 0$

$\phi(r) = \frac{K}{r} + C$ $K = U \cdot \frac{4\pi\epsilon_0}{a}$ $C = -U \cdot \frac{a}{b-a}$

$\oint_{\partial V} \vec{E} \cdot d\vec{A} = \iiint_V \rho_V(r) dV$

$Q = \iiint_V \rho_V dV$ $\rho_V = \frac{Q}{V}$

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Pot. einer Punktladung (Kreisscheibe)

$\phi(r) = \frac{1}{4\pi\epsilon_0} \cdot \int \frac{\rho_V(r')}{|r-r'|} dV'$

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Linienladung (Poi.) (nichtgleich unendlich)

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Spiegelung: Linienladung (r2 -> sp)

$\frac{\rho_L}{2\pi \cdot \epsilon} \cdot \ln\left(\frac{|r-r_2|}{|r-r_1|}\right)$

Stromdichte: $J = \int \vec{J} \cdot d\vec{A} = \delta_n \vec{H} \cdot d\vec{S}$

Feldstärke: $E = 1/K$

$R = \int_{r_1}^{r_2} \frac{1}{\kappa \cdot A} dr$ $dG = \frac{\kappa \cdot dA}{l}$

$\int_{R(0)}^R dR = \int_0^l \frac{dx}{\kappa \cdot A(x)}$ $R = \frac{l}{\kappa \cdot A}$

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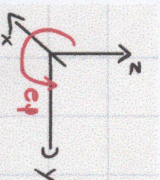
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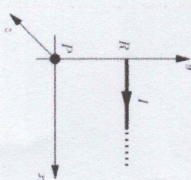
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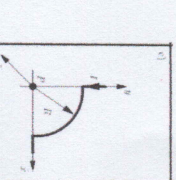


Spiegelung:
Ladung spiegeln (Q)
V2 > Punkt
 $\phi(r) = \frac{1}{4\pi\epsilon} \cdot \left(\frac{Q}{|r-r_1|} - \frac{Q}{|r-r_2|} \right)$
 $r = (x, y, z)$

Blot-Savart



$\vec{r} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$ $\vec{r} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$



$\vec{r} = \begin{pmatrix} R \cdot \cos(\varphi) \\ R \cdot \sin(\varphi) \\ 0 \end{pmatrix}$ $\int_{\pi/2}^0$

Linienladung: $E(r) = \frac{\rho_L}{2\pi\epsilon_0} \cdot \frac{1}{r}$ $W_e = \int \frac{1}{2} \cdot \epsilon \cdot E^2 dV = \frac{1}{2} \cdot C \cdot U^2$

Linienladung: $\phi(r) = \frac{\rho_L}{2\pi\epsilon} \cdot \ln\left(\frac{r_0}{|r-r'|}\right)$

Punktladung: $\phi(r) = \frac{Q}{4\pi\epsilon} \cdot \frac{1}{|r-r'|}$

r -> Wo messe ich?

Linie parallel zur Y-Achse: $\vec{r} = (x, 0, z)$

$\vec{r}' \rightarrow$ Wo ist die Ladung?

Linie bei z=0 z.B. $\vec{r}' = (0, 0, h)$

Wo ist das Pot. = 0?

Spiegelung benutz

Halbkreisbogen: dG ("rad" = Grad $\cdot \frac{\pi}{180}$)

$dA = b \cdot dr$ $l = \pi \cdot r$ $l = \alpha \cdot r$

Biot-Savart-Formel:

$\vec{B}(r) = \frac{\mu_0}{4\pi} \int \frac{d\vec{s} \times (r-r')}{|r-r'|^3}$ $d\vec{s} = \frac{dr}{dx}$

$\vec{r} = \begin{pmatrix} x \\ y \\ z \end{pmatrix}$ $\vec{r}' = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$ $d\vec{s} = \begin{pmatrix} 0 \\ 0 \\ dz \end{pmatrix}$

$\sqrt{x^2 + y^2} = \rho$

$\int \frac{dz}{\sqrt{a^2 + z^2}} = \frac{1}{a} \cdot \ln\left(\frac{a+z}{a-z}\right)$

Unendl. langer Leiter entlang z-Achse

rotch = j

$\oint_C \vec{H} \cdot d\vec{s} = \iint_A \vec{j} \cdot d\vec{A} = I$

$H \cdot 2 \cdot \pi \cdot r = I$

$dA = dA_1$ (Kreisscheibe)

$I_1 = \pi \cdot a^2 \cdot \epsilon_2$

Herleitung Punkt:

$\text{div } D = \rho_V$

$\epsilon \iint_S dA = \iint_S \rho_V dV$

differentielle Form	Satz	integrale Form
$\text{div } D = \rho_V$	$\rightarrow$ Gauss	$\oint_V D \cdot d\vec{A} = \iiint_V \rho_V dV$
$\text{div } \vec{B} = 0$	$\rightarrow$ Gauss	$\oint_V \vec{B} \cdot d\vec{A} = 0$
$\text{rot } \vec{E} = -\frac{\partial \vec{B}}{\partial t}$	$\rightarrow$ Stokes	$\oint_C \vec{E} \cdot d\vec{s} = - \iint_A \frac{\partial \vec{B}}{\partial t} \cdot d\vec{A}$
$\text{rot } \vec{H} = \vec{j} + \frac{\partial \vec{D}}{\partial t}$	$\rightarrow$ Stokes	$\oint_C \vec{H} \cdot d\vec{s} = \iint_A \vec{j} \cdot d\vec{A} + \frac{d}{dt} \left[\iint_A \vec{D} \cdot d\vec{A} \right]$

Laplace Gleichungen

Kartesisch: $\Delta\phi = A \cdot x + B$

Zylinder: $\Delta\phi = A \ln \rho + B$

Kugel: $\Delta\phi = A \cdot \varphi + B$

$\Delta(r) = \frac{1}{r} + B$

$\phi = -\int E d\varphi$

$E = -\nabla\phi$

Allgemein:

$H_2 = \int_A dA + \int_B dA$ $I_2 = \frac{I}{(c^2 - b^2)^{3/2}}$

$W_m = \int_V w_m dV$ $w_m = \frac{1}{2} \cdot B \cdot H = \frac{1}{2} \cdot \mu_0 \cdot H^2$

$W_m = \int_0^1 \int_0^{2\pi} \int_0^H \frac{1}{2} \cdot r \cdot dr \cdot d\varphi \cdot dz$

$\oint_D d\vec{A} = \int_V \rho_V \cdot \delta(x) dV$ $D_1 = D_2$ $2L_{\text{sg}} \cdot \rho_A \cdot A$

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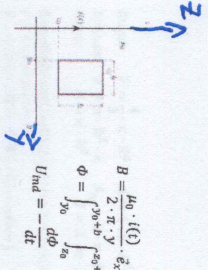
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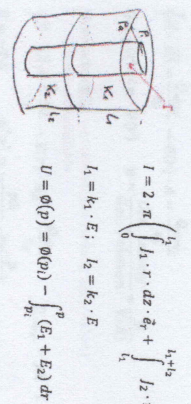
2 y Ebene



$$B = \frac{\mu_0}{2\pi} \cdot \frac{I}{r} \cdot \frac{dy}{r} = \frac{\mu_0 I}{2\pi} \int_{a-b}^a \frac{1}{r^2} dy$$

$$\Phi = \int_{y_0}^{y_0+b} \int_{x_0}^{x_0+a} B \cdot dA_x$$

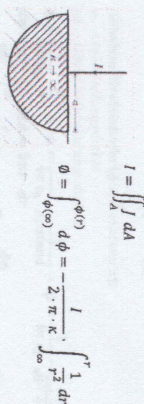
$$U_{ind} = - \frac{d\Phi}{dt}$$



$$I = 2 \cdot \pi \cdot \left(\int_{r_1}^{r_2} I_1 \cdot r \cdot dz \cdot \vec{e}_r + \int_{r_1}^{r_2} I_2 \cdot r \cdot dz \cdot \vec{e}_r \right)$$

$$I_1 = k_1 \cdot E, \quad I_2 = k_2 \cdot E$$

$$U = \Phi(r) = \Phi(r_1) - \int_{r_1}^r (E_1 + E_2) dr$$



$$I = \iint_A j \, dA$$

$$\Phi = \int_{\phi(\infty)}^{\phi(r)} d\phi = - \frac{1}{2\pi \cdot \pi \cdot R} \cdot \int_{\infty}^r \frac{1}{r^2} dr$$

Kreisscheibe:

$$\phi(r) = \frac{1}{4 \cdot \pi \cdot \epsilon_0} \cdot \int_A \frac{\rho(r')}{|r - r'|} dA$$

$$\phi(r) = \frac{\rho r}{4 \cdot \pi \cdot \epsilon_0} \cdot \int_0^{2\pi} \int_0^R \frac{1}{\sqrt{r^2 + r'^2 - 2rr' \cos \alpha}} \cdot r' \cdot d\alpha \cdot dr' \cdot \vec{e}_z$$

Bestimme das elektrische Potential $\Phi(r)$ einer homogen geladenen Kugel (Radius R, Ladungsdichte ρ , Permittivität ϵ im gesamten Raum mittels Poisson- und Laplace-Gleichung unter der Randbedingung $\Phi(r \rightarrow \infty) = 0$.

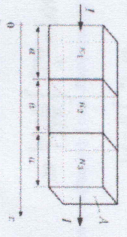
$$\frac{d}{dr} \left(r^2 \frac{d\Phi}{dr} \right) = 0$$

$$r^2 \cdot \frac{d\Phi}{dr} = C_1$$

$$\frac{d\Phi}{dr} = \frac{C_1}{r^2}$$

$$\Phi = - \frac{C_1}{r} + C_2$$

$$\Phi(r) = -C_1 \cdot \frac{1}{r} + C_2$$

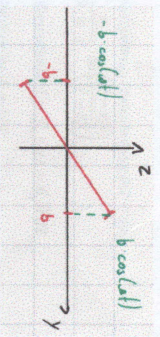


$$I = \int j \, dA$$

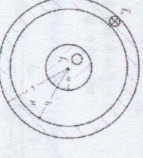
$$j \rightarrow \text{konst.}$$

$$E_1 = \frac{A \cdot K}{A \cdot K} \cdot E_2 = \dots$$

$$\Phi_1(x) = - \left(\int_{x_1}^x E_1 dx + \int_{x_2}^x E_2 dx + \int_{x_3}^{x_0} E_3 dx \right)$$



$$U_{ind} = - \frac{d}{dt} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \vec{B} \cdot d\vec{A}$$



$$\oint \vec{H} \cdot d\vec{s} = H \cdot 2\pi r$$

$$I_{enc} = j \cdot A = I \cdot \frac{\rho^2}{a^2}$$

$$\vec{H}(r) = \frac{I \cdot \rho}{2\pi a^2} \vec{e}_\phi, \quad 0 < \rho \leq a$$

$$\vec{H}(r) = \frac{I}{2\pi \rho} \vec{e}_\phi, \quad a < \rho \leq b$$

$$\vec{H}(r) = \frac{I}{2\pi \rho} \left(1 - \frac{\rho^2 - b^2}{a^2 - b^2} \right) \vec{e}_\phi, \quad b < \rho \leq c$$

Punktladung: (L=r)

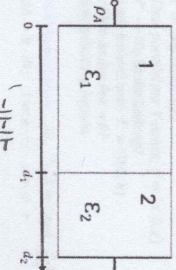
$$\oint \vec{E} \cdot d\vec{s} = \frac{Q}{\epsilon_0} = \frac{\rho \cdot V}{\epsilon_0} = \frac{\rho \cdot \frac{4}{3}\pi r^3}{\epsilon_0}$$

Umladung:

$$\oint \vec{E} \cdot d\vec{s} = \frac{Q}{\epsilon_0} = \frac{\rho \cdot V}{\epsilon_0} = \frac{\rho \cdot \frac{4}{3}\pi r^3}{\epsilon_0}$$

Spiegelung Umladung

$$\Phi(r) = \frac{\rho_1}{2 \cdot \pi \cdot \epsilon} \cdot \ln \left(\frac{r - r_1}{r - r_2} \right)$$



$$\oint \vec{E} \cdot d\vec{s} = \int \rho_1 \, dA \quad D = E \cdot \epsilon$$

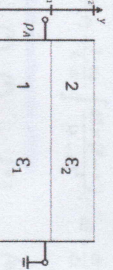
$$\vec{E} = \frac{\rho_1}{\epsilon_1} \cdot \vec{e}_x \quad ; \quad (\vec{E}_1 \neq \vec{E}_2)$$

$$\vec{E} = -\nabla \Phi = -\frac{\partial \Phi}{\partial x} \vec{e}_x$$

$$\Phi_1(x) = \int -\frac{\rho_1}{\epsilon_1} dx + C_1 = -\frac{\rho_1}{\epsilon_1} x + \frac{\rho_1}{\epsilon_1} a_1$$

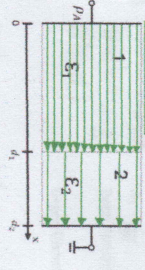
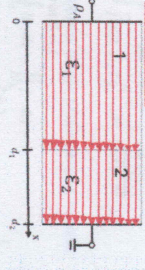
$$\Phi_2(x) = \int -\frac{\rho_2}{\epsilon_2} dx + C_2 = -\frac{\rho_2}{\epsilon_2} x + \frac{\rho_2}{\epsilon_2} a_2$$

$$\Phi(x) = \Phi_1(x) - \Phi_2(x)$$



$$E = \frac{\rho_1 \cdot A}{\epsilon A_1 + 2\epsilon A_2} = \frac{\rho_1 \cdot a_2}{\epsilon A_1 + 2\epsilon \cdot (a_2 - a_1) \cdot (x - d)}$$

$$\Phi(x) = - \frac{\rho_1 \cdot a_2}{\epsilon A_1 + 2\epsilon \cdot (a_2 - a_1) \cdot (x - d)}$$



$$C = \frac{\epsilon_1 \cdot A_1 + \epsilon_2 \cdot A_2}{d}$$

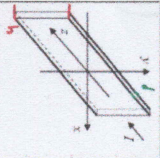
Aufgabe 4: Kugelskondensator

An einem Kugelskondensator (Innen leitende Elektroden) gilt dem Innerenradius a, dem Außenradius b und einem Dielektrikum mit der Permittivität ϵ wird eine Gleichspannung $U > 0$ angelegt, wobei die Außenkugelelektrode geerdet ist. Berechnen Sie das E-Feld $E(r)$ im gesamten Raum sowie die Kapazität C des Kondensators.

$$E(r) = \begin{cases} \frac{U}{a} \frac{a^2}{r^2}, & a \leq r \leq b \\ 0, & \text{sonst} \end{cases} \quad C = 4\pi \epsilon \frac{a \cdot b}{b - a}$$

$$\Phi(r) = \frac{A}{r} + B \quad \Phi(r) = U = \frac{b - r \cdot a}{b - a} \quad \vec{E}(r) = -\nabla \Phi = \frac{U}{r^2} \vec{e}_r \cdot \vec{e}_r = \vec{E}$$

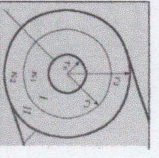
$$Q = \oint_A D \cdot dA = Q = E \cdot 4 \cdot \pi \cdot r^2$$



$$\frac{dl}{r} = \frac{1}{r} \cdot dy \cdot \vec{e}_y; \quad r = x \cdot \vec{e}_x$$

$$r' = y' \cdot \vec{e}_y; \quad \int -b/2 \rightarrow b/2$$

$$dH = \frac{1}{2 \cdot \pi \cdot r^2} (dl \times (r - r'))$$



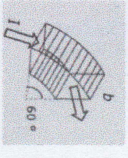
$$\frac{I_1}{\epsilon_1} = \frac{I_2}{\epsilon_2} \quad E = \frac{I}{2\pi r L \cdot \epsilon}$$

$$\Phi_1 = - \int_{r_1}^{r_2} E_1 \, dr$$

$$\Phi_{II} = \Phi_I - \int_{r_1}^{r_2} \frac{I}{2\pi r L \cdot \epsilon_2} \, dr$$

$$U = \Phi_{II} - \Phi_{II}$$

Wie groß ist der Widerstand der Anordnung (Stromfluss I) bei einem Außenradius ra, einem Innenradius ri, einer Leitfähigkeit κ , einer Dicke b und einem Winkel von 60°?



$$R = \int_{r_1}^{r_2} \frac{1}{\kappa \cdot b \cdot \alpha \cdot r} \, dr = \frac{1}{3} \ln \frac{r_2}{r_1}$$

gehört κ (Leitfähigkeit) desto kleiner E-Feld

$E = d$ Feldstärke

$D = el$ Flussdichte

$H = \text{magn. Feldstärke}$

$B = \text{magn. Flussdichte}$

$j = \text{Strom dichte}$

$\rho = \text{Raumladungsdichte}$

$\epsilon = \text{Permittivität (für E-Feld)}$

$\mu = \text{Permeabilität (für B-Feld)}$

$\kappa = \text{Leitfähigkeit}$

$$O = \epsilon E$$

$$B = \mu \cdot H$$

$$j = k \cdot E$$

$$\text{Mittelwert} \rightarrow \text{Vektorprodukt} = 0$$

$$\text{Radial frei} \rightarrow \text{divergenz} = 0$$

$$e^0 = 1 \quad e^d = \infty \quad e^{-d} = 0$$

$$\ln(1) = 0 \quad \ln(0) = -\infty$$

$$2\pi = 360^\circ \quad \pi = 180^\circ \quad \frac{\pi}{2} = 90^\circ$$

$$\frac{\pi}{3} = 60^\circ \quad \frac{\pi}{4} = 45^\circ \quad \frac{\pi}{6} = 30^\circ$$

$$\sin \rightarrow 0 \rightarrow 1 \quad \cos \rightarrow 1 \rightarrow 0$$

groß $\rightarrow$ Ladungen sind außen

verändertes Magnetfeld erzeugt E-Feld $\text{rot } E = -\frac{\partial B}{\partial t}$

Elektrostatik $\rightarrow$ keine Ströme $\oint D \cdot dA = Q$

Magnetostatik $\rightarrow$ zeitlich konst. Ströme, keine Ladungsdichte

die sich bewegen $\oint H \cdot dA = I$

von Magnetfeldänderung wird Spannung induziert

$\text{Uind} = - \frac{d\Phi}{dt}$

Skin-Effekt $\rightarrow$ hohe Frequenz - Strom fließt nur nach außen Oberfl. von Leiter

$\delta = \sqrt{\frac{2}{\omega \mu \kappa}}$

$\omega = 2\pi f$